

Nowshin Farhana

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SUMMARY

Highly motivated AI/ML enthusiast passionate about Artificial Intelligence and Machine Learning. Practical experience in building ML models, analyzing datasets, and implementing NLP-based projects. Strong foundation in Python and core ML concepts, with a continuous learning mindset and enthusiasm to contribute to impactful AI solutions. Actively seeking a Summer Internship in AI/ML to gain hands-on industry experience and further develop practical machine learning skills.

EDUCATION

Rajiv Gandhi University of Knowledge Technologies, Nuzvid <i>Bachelor of Technology (B.Tech) in Computer Science Engineering</i> CGPA: 9.0	2023 – 2027
Rajiv Gandhi University of Knowledge Technologies, Nuzvid <i>Pre-University Course (PUC)</i> CGPA: 9.92	2021 – 2023

TECHNICAL SKILLS

Languages: Python, JavaScript, C++, C
Frontend & Backend: React.js, HTML, CSS, Bootstrap, Node.js, Streamlit
AI/ML: Machine Learning, Deep Learning, TensorFlow, OpenCV, Pandas, NumPy
Blockchain & Security: Ethereum, Solidity, Smart Contracts, MetaMask, IPFS
Databases & Tools: MongoDB, Git, GitHub, VS Code,

EXPERIENCE

Parabola9 AI/ML Intern	Dec 2024 – April 2025
- Worked on real-time operational datasets, performing structured data labeling and annotation to support machine learning workflows. - Ensured data quality and consistency through validation and manual review processes under time-sensitive scenarios. - Prepared and organized labeled datasets to improve downstream model training efficiency and reliability. - Collaborated with cross-functional teams to maintain accuracy standards and streamline dataset preparation processes.	
DevSecEng Ops — Web Development & Python Intern	March 2024 – October 2024
- Developed responsive web interfaces and contributed to frontend feature implementation for internal web applications. - Built Python-based web scraping pipelines to extract, clean, and process structured data from multiple online sources. - Collaborated with development teams to optimize scraping scripts, handle dynamic websites, and maintain data accuracy.	

PROJECTS

AI-Powered Dynamic Route Rationalization System (Smart India Hackathon)
Tech Stack: Python, PyTorch, Pandas, NumPy, CLSTM, Ant Colony Optimization, Kafka, Google Maps API. - Spearheaded the development of a machine learning-driven dynamic route optimization system to prevent bus bunching and improve real-time urban transport efficiency. - Designed and trained a hybrid CNN-LSTM (CLSTM) model for spatial-temporal traffic analysis, achieving 91% delay prediction accuracy. - Implemented Ant Colony Optimization (ACO), reducing predicted route delays by 18–22%. - Validated performance using real-time streaming data to ensure scalability and robustness.
Musentia — Facial Emotion-Based Music Recommendation System
Tech Stack: Python, TensorFlow, OpenCV, MediaPipe, CNN, Streamlit, NumPy, Pandas. - Built an end-to-end AI system for real-time facial emotion detection and personalized music recommendation.

- Trained a CNN-based emotion classifier integrated with MediaPipe, achieving **87% accuracy**.
- Designed a modular pipeline linking emotion recognition with a dynamic recommendation engine.
- Developed an interactive Streamlit interface for real-time emotion capture and playlist generation.

Blockchain-Based Secure Healthcare Data Management System with AI Integration

- **Tech Stack:** React.js, Node.js, Express.js, MongoDB, Solidity, Ethereum, IPFS, MetaMask, XGBoost, Random Forest, Logistic Regression.
- Developed a decentralized healthcare platform enabling secure patient-owned medical record management using Ethereum blockchain, IPFS, and smart contracts.
- Implemented end-to-end encrypted medical record storage and sharing using AES-GCM encryption, ECIES cryptography, and NFT-based access authorization.
- Built a full-stack architecture supporting decentralized authentication and secure healthcare workflows.
- Integrated an AI-powered disease prediction module using XGBoost, Logistic Regression, Random Forest, and ensemble learning for intelligent healthcare insights.
- Designed transparent audit tracking, doctor verification, and blockchain-based access revocation to ensure secure and accountable healthcare collaboration.

ACHIEVEMENTS

- **Winner – Smart India Hackathon (SIH) 2024**, selected among the top teams nationwide under AICTE evaluation.
- **Finalist – Aaavishkar Hackathon**, recognized for innovative AI-driven solution development.
- **NPTEL Certification – Search Methods for Problem Solving in Artificial Intelligence IIT Madras**, secured **76%**.
- **Summer Analytics '25 – Data Science and Machine Learning Program** conducted by Data Analytics club of IIT Guwahati.

EXTRA-CURRICULAR ACTIVITIES

- **InfoTechies Club Member** – Participated in technical discussions, project ideation, and hackathons focused on innovative software solutions.
- **Campus Ambassador** – Represented my campus for EXERGY (Chemical Engineering Fest), coordinating outreach activities and promoting event participation among students.